

Trees

- 11.1 Introduction to Trees
- 11.2 Applications of Trees
- 11.3 Tree Traversal
- 11.4 Spanning Trees
- 11.5 Minimum Spanning Trees

A connected graph that contains no simple circuits is called a tree. Trees were used as long ago as 1857, when the English mathematician Arthur Cayley used them to count certain types of chemical compounds. Since that time, trees have been employed to solve problems in a wide variety of disciplines, as the examples in this chapter will show.

Trees are particularly useful in computer science, where they are employed in a wide range of algorithms. For instance, trees are used to construct efficient algorithms for locating items in a list. They can be used in algorithms, such as Huffman coding, that construct efficient codes saving costs in data transmission and storage. Trees can be used to study games such as checkers and chess and can help determine winning strategies for playing these games. Trees can be used to model procedures carried out using a sequence of decisions. Constructing these models can help determine the computational complexity of algorithms based on a sequence of decisions, such as sorting algorithms.

Procedures for building trees containing every vertex of a graph, including depth-first search and breadth-first search, can be used to systematically explore the vertices of a graph. Exploring the vertices of a graph via depth-first search, also known as backtracking, allows for the systematic search for solutions to a wide variety of problems, such as determining how eight queens can be placed on a chessboard so that no queen can attack another.

We can assign weights to the edges of a tree to model many problems. For example, using weighted trees we can develop algorithms to construct networks containing the least expensive set of telephone lines linking different network nodes.

11.1 Introduction to Trees



In Chapter 10 we showed how graphs can be used to model and solve many problems. In this chapter we will focus on a particular type of graph called a **tree**, so named because such graphs resemble trees. For example, *family trees* are graphs that represent genealogical charts. Family trees use vertices to represent the members of a family and edges to represent parent–child relationships. The family tree of the male members of the Bernoulli family of Swiss mathematicians is shown in Figure 1. The undirected graph representing a family tree (restricted to people of just one gender and with no inbreeding) is an example of a tree.

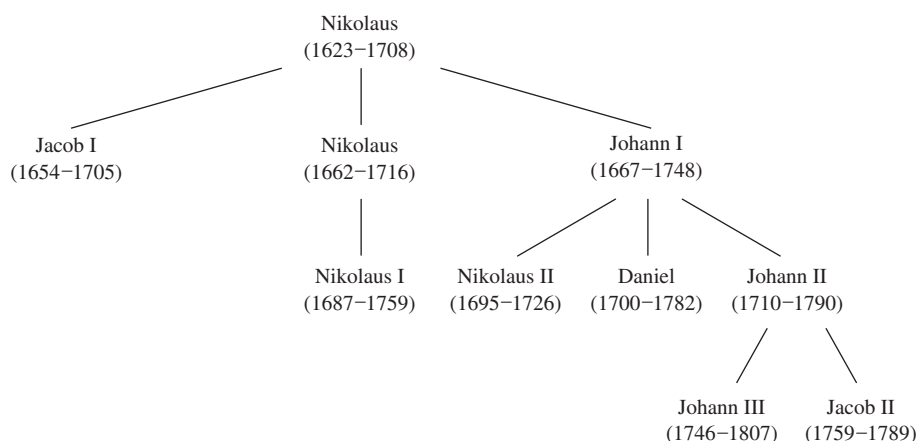


FIGURE 1 The Bernoulli family of mathematicians.

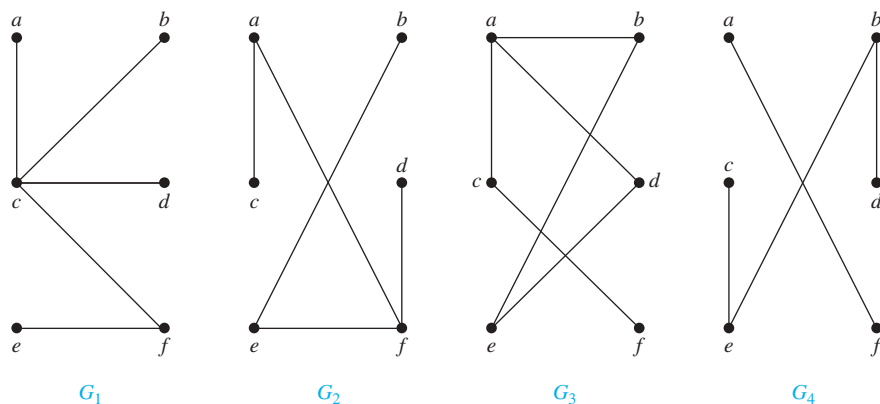


FIGURE 2 Examples of trees and graphs that are not trees.

Definition 1

A *tree* is a connected undirected graph with no simple circuits.

Because a tree cannot have a simple circuit, a tree cannot contain multiple edges or loops. Therefore any tree must be a simple graph.

EXAMPLE 1

Which of the graphs shown in Figure 2 are trees?

Solution: G_1 and G_2 are trees, because both are connected graphs with no simple circuits. G_3 is not a tree because e, b, a, d, e is a simple circuit in this graph. Finally, G_4 is not a tree because it is not connected. ▶

Any connected graph that contains no simple circuits is a tree. What about graphs containing no simple circuits that are not necessarily connected? These graphs are called **forests** and have the property that each of their connected components is a tree. Figure 3 displays a forest. Note that the graph G_4 in Figure 2 is also a forest. The graph in Figure 3 is a forest of three trees, while G_4 in Figure 2 is a forest of two trees.

Trees are often defined as undirected graphs with the property that there is a unique simple path between every pair of vertices. Theorem 1 shows that this alternative definition is equivalent to our definition.

THEOREM 1

An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

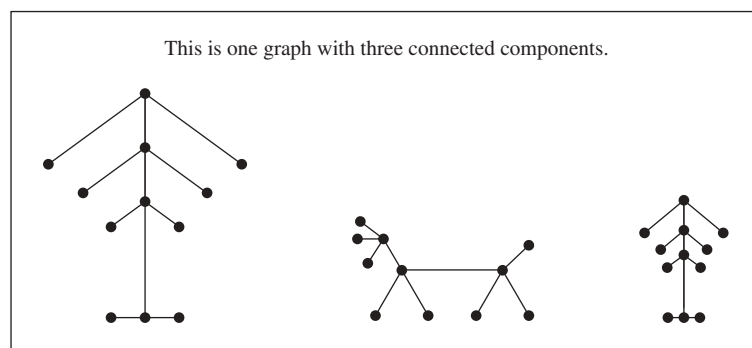


FIGURE 3 Example of a forest.

Proof: First assume that T is a tree. Then T is a connected graph with no simple circuits. Let x and y be two vertices of T . Because T is connected, by Theorem 1 of Section 10.4 there is a simple path between x and y . Moreover, this path must be unique, for if there were a second such path, the path formed by combining the first path from x to y followed by the path from y to x obtained by reversing the order of the second path from x to y would form a circuit. This implies, using Exercise 59 of Section 10.4, that there is a simple circuit in T . Hence, there is a unique simple path between any two vertices of a tree.

Now assume that there is a unique simple path between any two vertices of a graph T . Then T is connected, because there is a path between any two of its vertices. Furthermore, T can have no simple circuits. To see that this is true, suppose T had a simple circuit that contained the vertices x and y . Then there would be two simple paths between x and y , because the simple circuit is made up of a simple path from x to y and a second simple path from y to x . Hence, a graph with a unique simple path between any two vertices is a tree. $\triangleleft$

11.1.1 Rooted Trees

In many applications of trees, a particular vertex of a tree is designated as the **root**. Once we specify a root, we can assign a direction to each edge as follows. Because there is a unique path from the root to each vertex of the graph (by Theorem 1), we direct each edge away from the root. Thus, a tree together with its root produces a directed graph called a **rooted tree**.

Definition 2

A *rooted tree* is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

Rooted trees can also be defined recursively. Refer to Section 5.3 to see how this can be done. We can change an unrooted tree into a rooted tree by choosing any vertex as the root. Note that different choices of the root produce different rooted trees. For instance, Figure 4 displays the rooted trees formed by designating a to be the root and c to be the root, respectively, in the tree T . We usually draw a rooted tree with its root at the top of the graph. The arrows indicating the directions of the edges in a rooted tree can be omitted, because the choice of root determines the directions of the edges.

The terminology for trees has botanical and genealogical origins. Suppose that T is a rooted tree. If v is a vertex in T other than the root, the **parent** of v is the unique vertex u such that there is a directed edge from u to v (the reader should show that such a vertex is unique). When u is the parent of v , v is called a **child** of u . Vertices with the same parent are called **siblings**. The **ancestors** of a vertex other than the root are the vertices in the path from the root to this vertex, excluding the vertex itself and including the root (that is, its parent, its parent's parent, and so on, until the root is reached). The **descendants** of a vertex v are those vertices that have v as an ancestor. A vertex of a rooted tree is called a **leaf** if it has no children. Vertices that have children are called **internal vertices**. The root is an internal vertex unless it is the only vertex in the graph, in which case it is a leaf.

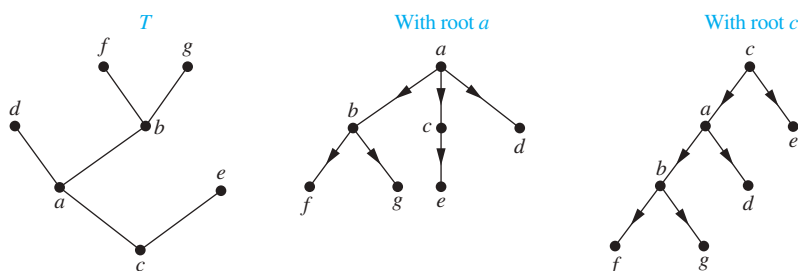
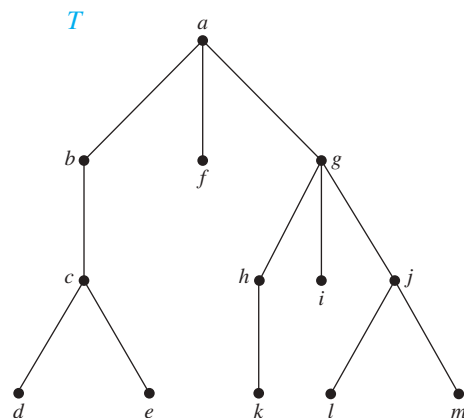
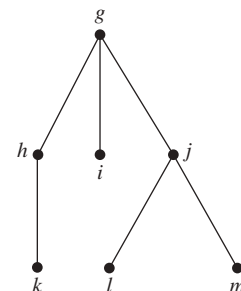


FIGURE 4 A tree and rooted trees formed by designating two different roots.

FIGURE 5 A rooted tree T .FIGURE 6 The subtree rooted at g .

If a is a vertex in a tree, the **subtree** with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

EXAMPLE 2 In the rooted tree T (with root a) shown in Figure 5, find the parent of c , the children of g , the siblings of h , all ancestors of e , all descendants of b , all internal vertices, and all leaves. What is the subtree rooted at g ?

Extra Examples ➤

Solution: The parent of c is b . The children of g are h , i , and j . The siblings of h are i and j . The ancestors of e are c , b , and a . The descendants of b are c , d , and e . The internal vertices are a , b , c , g , h , and j . The leaves are d , e , f , i , k , l , and m . The subtree rooted at g is shown in Figure 6. ➤

Rooted trees with the property that all of their internal vertices have the same number of children are used in many different applications. Later in this chapter we will use such trees to study problems involving searching, sorting, and coding.

Definition 3

A rooted tree is called an *m -ary tree* if every internal vertex has no more than m children. The tree is called a *full m -ary tree* if every internal vertex has exactly m children. An m -ary tree with $m = 2$ is called a *binary tree*.

Links ➤

EXAMPLE 3 Are the rooted trees in Figure 7 full m -ary trees for some positive integer m ?

Solution: T_1 is a full binary tree because each of its internal vertices has two children. T_2 is a full 3-ary tree because each of its internal vertices has three children. In T_3 each internal vertex

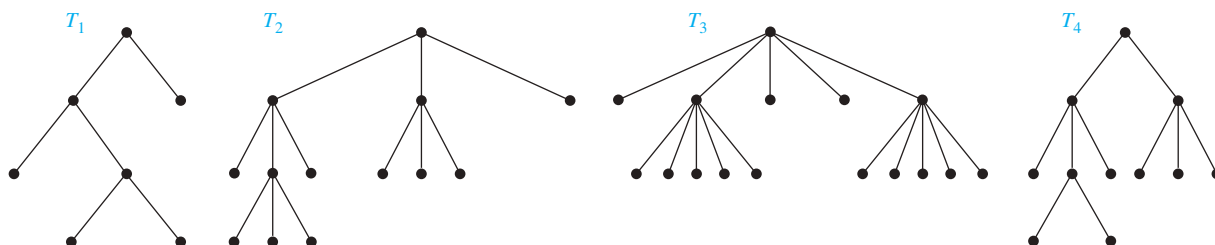


FIGURE 7 Four rooted trees.

has five children, so T_3 is a full 5-ary tree. T_4 is not a full m -ary tree for any m because some of its internal vertices have two children and others have three children. ◀

ORDERED ROOTED TREES An **ordered rooted tree** is a rooted tree where the children of each internal vertex are ordered. Ordered rooted trees are drawn so that the children of each internal vertex are shown in order from left to right. Note that a representation of a rooted tree in the conventional way determines an ordering for its edges. We will use such orderings of edges in drawings without explicitly mentioning that we are considering a rooted tree to be ordered.

In an ordered binary tree (usually called just a **binary tree**), if an internal vertex has two children, the first child is called the **left child** and the second child is called the **right child**. The tree rooted at the left child of a vertex is called the **left subtree** of this vertex, and the tree rooted at the right child of a vertex is called the **right subtree** of the vertex. The reader should note that for some applications every vertex of a binary tree, other than the root, is designated as a right or a left child of its parent. This is done even when some vertices have only one child. We will make such designations whenever it is necessary, but not otherwise.

Ordered rooted trees can be defined recursively. Binary trees, a type of ordered rooted trees, were defined this way in Section 5.3.

EXAMPLE 4 What are the left and right children of d in the binary tree T shown in Figure 8(a) (where the order is that implied by the drawing)? What are the left and right subtrees of c ?

Solution: The left child of d is f and the right child is g . We show the left and right subtrees of c in Figures 8(b) and 8(c), respectively. ◀

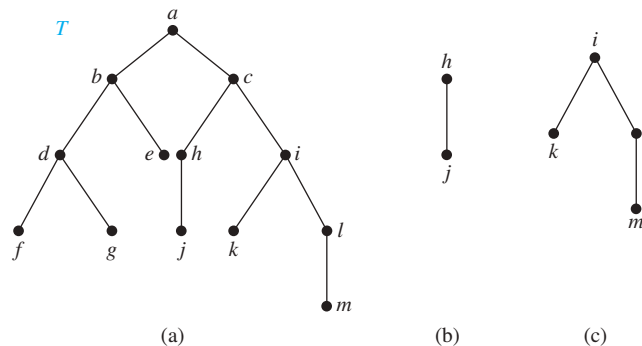


FIGURE 8 A binary tree T and left and right subtrees of the vertex c .

Just as in the case of graphs, there is no standard terminology used to describe trees, rooted trees, ordered rooted trees, and binary trees. This nonstandard terminology occurs because trees are used extensively throughout computer science, which is a relatively young field. The reader should carefully check meanings given to terms dealing with trees whenever they occur.

11.1.2 Trees as Models

Trees are used as models in such diverse areas as computer science, chemistry, geology, botany, and psychology. We will describe a variety of such models based on trees.

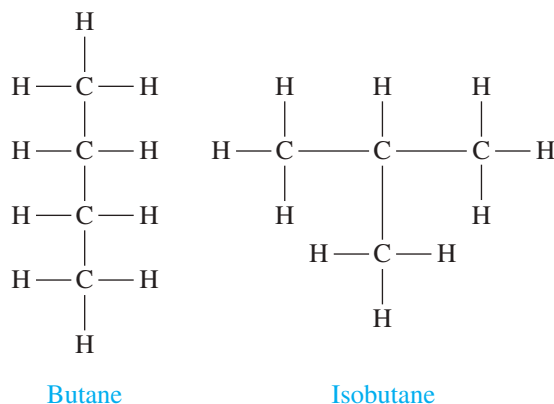


FIGURE 9 The two isomers of butane.

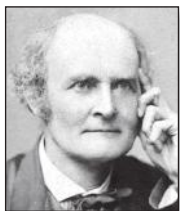
EXAMPLE 5 Saturated Hydrocarbons and Trees Graphs can be used to represent molecules, where atoms are represented by vertices and bonds between them by edges. The English mathematician Arthur Cayley discovered trees in 1857 when he was trying to enumerate the isomers of compounds of the form C_nH_{2n+2} , which are called *saturated hydrocarbons*. (Isomers represent compounds with the same chemical formula but different chemical properties.)

In graph models of saturated hydrocarbons, each carbon atom is represented by a vertex of degree 4, and each hydrogen atom is represented by a vertex of degree 1. There are $3n + 2$ vertices in a graph representing a compound of the form C_nH_{2n+2} . The number of edges in such a graph is half the sum of the degrees of the vertices. Hence, there are $(4n + 2n + 2)/2 = 3n + 1$ edges in this graph. Because the graph is connected and the number of edges is one less than the number of vertices, it must be a tree (see Exercise 15).

The nonisomorphic trees with n vertices of degree 4 and $2n + 2$ of degree 1 represent the different isomers of C_nH_{2n+2} . For instance, when $n = 4$, there are exactly two nonisomorphic trees of this type (the reader should verify this). Hence, there are exactly two different isomers of C_4H_{10} . Their structures are displayed in Figure 9. These two isomers are called butane and isobutane (also known as i-butane or methylpropane). ◀

EXAMPLE 6 Representing Organizations The structure of a large organization can be modeled using a rooted tree. Each vertex in this tree represents a position in the organization. An edge from one vertex to another indicates that the person represented by the initial vertex is the (direct) boss of the person represented by the terminal vertex. The graph shown in Figure 10 displays

Links



©Paul Fearn/Alamy Stock Photo

ARTHUR CAYLEY (1821–1895) Arthur Cayley, the son of a merchant, displayed his mathematical talents at an early age with amazing skill in numerical calculations. Cayley entered Trinity College, Cambridge, when he was 17. While in college he developed a passion for reading novels. Cayley excelled at Cambridge and was elected to a 3-year appointment as Fellow of Trinity and assistant tutor. During this time Cayley began his study of n -dimensional geometry and made a variety of contributions to geometry and to analysis. He also developed an interest in mountaineering, which he enjoyed during vacations in Switzerland. Because no position as a mathematician was available to him, Cayley left Cambridge, entering the legal profession and gaining admittance to the bar in 1849. Although Cayley limited his legal work to be able to continue his mathematics research, he developed a reputation as a legal specialist. During his legal career he was able to write more than 300 mathematical papers. In 1863 Cambridge University established a new post in mathematics and offered it to Cayley. He took this job, even though it paid less money than he made as a lawyer.

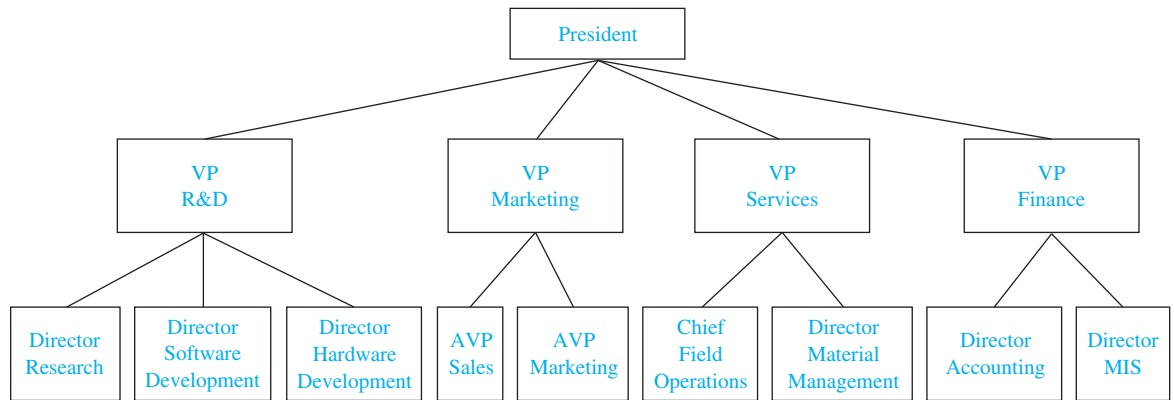


FIGURE 10 An organizational tree for a computer company.

such a tree. In the organization represented by this tree, the Director of Hardware Development works directly for the Vice President of R&D. The root of this tree is the vertex representing the President of the organization. ◀

EXAMPLE 7 Computer File Systems Files in computer memory can be organized into directories. A directory can contain both files and subdirectories. The root directory contains the entire file system. Thus, a file system may be represented by a rooted tree, where the root represents the root directory, internal vertices represent subdirectories, and leaves represent ordinary files or empty directories. One such file system is shown in Figure 11. In this system, the file *chr* is in the directory *rje*. (Note that links to files where the same file may have more than one pathname can lead to circuits in computer file systems.) ◀

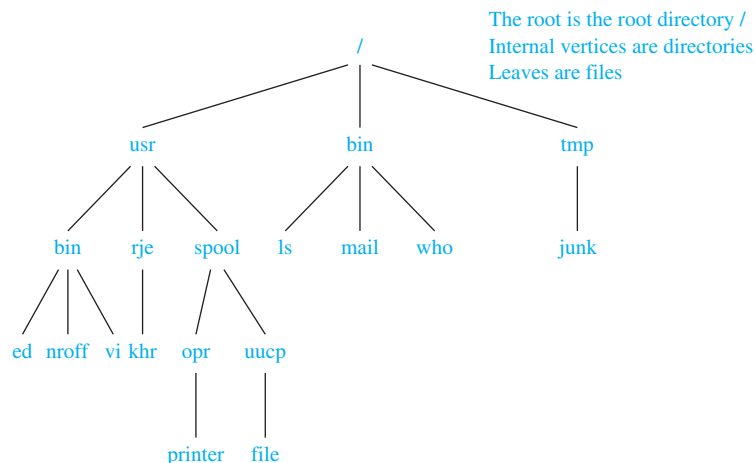


FIGURE 11 A computer file system.

EXAMPLE 8 Tree-Connected Parallel Processors In Example 17 of Section 10.2 we described several interconnection networks for parallel processing. A **tree-connected network** is another important way to interconnect processors. The graph representing such a network is a complete binary tree, that is, a full binary tree where every root is at the same level. Such a network interconnects

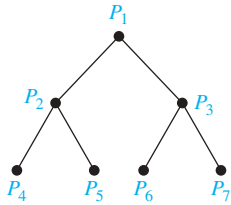


FIGURE 12 A tree-connected network of seven processors.

$n = 2^k - 1$ processors, where k is a positive integer. A processor represented by the vertex v that is not a root or a leaf has three two-way connections—one to the processor represented by the parent of v and two to the processors represented by the two children of v . The processor represented by the root has two two-way connections to the processors represented by its two children. A processor represented by a leaf v has a single two-way connection to the parent of v . We display a tree-connected network with seven processors in Figure 12.

We now illustrate how a tree-connected network can be used for parallel computation. In particular, we show how the processors in Figure 12 can be used to add eight numbers, using three steps. In the first step, we add x_1 and x_2 using P_4 , x_3 and x_4 using P_5 , x_5 and x_6 using P_6 , and x_7 and x_8 using P_7 . In the second step, we add $x_1 + x_2$ and $x_3 + x_4$ using P_2 and $x_5 + x_6$ and $x_7 + x_8$ using P_3 . Finally, in the third step, we add $x_1 + x_2 + x_3 + x_4$ and $x_5 + x_6 + x_7 + x_8$ using P_1 . The three steps used to add eight numbers compares favorably to the seven steps required to add eight numbers serially, where the steps are the addition of one number to the sum of the previous numbers in the list. ◀

11.1.3 Properties of Trees

We will often need results relating the numbers of edges and vertices of various types in trees.

THEOREM 2

A tree with n vertices has $n - 1$ edges.

Links

Proof: We will use mathematical induction to prove this theorem. Note that for all the trees here we can choose a root and consider the tree rooted.

BASIS STEP: When $n = 1$, a tree with $n = 1$ vertex has no edges. It follows that the theorem is true for $n = 1$.

INDUCTIVE STEP: The inductive hypothesis states that every tree with k vertices has $k - 1$ edges, where k is a positive integer. Suppose that a tree T has $k + 1$ vertices and that v is a leaf of T (which must exist because the tree is finite), and let w be the parent of v . Removing from T the vertex v and the edge connecting w to v produces a tree T' with k vertices, because the resulting graph is still connected and has no simple circuits. By the inductive hypothesis, T' has $k - 1$ edges. It follows that T has k edges because it has one more edge than T' , the edge connecting v and w . This completes the inductive step. ◀

Recall that a tree is a connected undirected graph with no simple circuits. So, when G is an undirected graph with n vertices, Theorem 2 tells us that the two conditions (i) G is connected and (ii) G has no simple circuits, imply (iii) G has $n - 1$ edges. Also, when (i) and (iii) hold, then (ii) must also hold, and when (ii) and (iii) hold, (i) must also hold. That is, if G is connected and G has $n - 1$ edges, then G has no simple circuits, so that G is a tree (see Exercise 15(a)), and if G has no simple circuits and G has $n - 1$ edges, then G is connected, and so is a tree (see Exercise 15(b)). Consequently, when two of (i), (ii), and (iii) hold, the third condition must also hold, and G must be a tree.

COUNTING VERTICES IN FULL m -ARY TREES The number of vertices in a full m -ary tree with a specified number of internal vertices is determined, as Theorem 3 shows. As in Theorem 2, we will use n to denote the number of vertices in a tree.

THEOREM 3

A full m -ary tree with i internal vertices contains $n = mi + 1$ vertices.

Proof: Every vertex, except the root, is the child of an internal vertex. Because each of the i internal vertices has m children, there are mi vertices in the tree other than the root. Therefore, the tree contains $n = mi + 1$ vertices. ◀

Suppose that T is a full m -ary tree. Let i be the number of internal vertices and l the number of leaves in this tree. Once one of n , i , and l is known, the other two quantities are determined. Theorem 4 explains how to find the other two quantities from the one that is known.

THEOREM 4

A full m -ary tree with

- (i) n vertices has $i = (n - 1)/m$ internal vertices and $l = [(m - 1)n + 1]/m$ leaves,
- (ii) i internal vertices has $n = mi + 1$ vertices and $l = (m - 1)i + 1$ leaves,
- (iii) l leaves has $n = (ml - 1)/(m - 1)$ vertices and $i = (l - 1)/(m - 1)$ internal vertices.

Proof: Let n represent the number of vertices, i the number of internal vertices, and l the number of leaves. The three parts of the theorem can all be proved using the equality given in Theorem 3, that is, $n = mi + 1$, together with the equality $n = l + i$, which is true because each vertex is either a leaf or an internal vertex. We will prove part (i) here. The proofs of parts (ii) and (iii) are left as exercises for the reader.

Solving for i in $n = mi + 1$ gives $i = (n - 1)/m$. Then inserting this expression for i into the equation $n = l + i$ shows that $l = n - i = n - (n - 1)/m = [(m - 1)n + 1]/m$. ◀

Example 9 illustrates how Theorem 4 can be used.

EXAMPLE 9

Suppose that someone starts a chain letter. Each person who receives the letter is asked to send it on to four other people. Some people do this, but others do not send any letters. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? How many people sent out the letter?

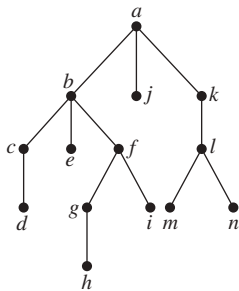


FIGURE 13 A rooted tree.

Solution: The chain letter can be represented using a 4-ary tree. The internal vertices correspond to people who sent out the letter, and the leaves correspond to people who did not send it out. Because 100 people did not send out the letter, the number of leaves in this rooted tree is $l = 100$. Hence, part (iii) of Theorem 4 shows that the number of people who have seen the letter is $n = (4 \cdot 100 - 1)/(4 - 1) = 133$. Also, the number of internal vertices is $133 - 100 = 33$, so 33 people sent out the letter. ◀

BALANCED m -ARY TREES It is often desirable to use rooted trees that are “balanced” so that the subtrees at each vertex contain paths of approximately the same length. Some definitions will make this concept clear. The **level** of a vertex v in a rooted tree is the length of the unique path from the root to this vertex. The level of the root is defined to be zero. The **height** of a rooted tree is the maximum of the levels of vertices. In other words, the height of a rooted tree is the length of the longest path from the root to any vertex.

EXAMPLE 10

Find the level of each vertex in the rooted tree shown in Figure 13. What is the height of this tree?

Solution: The root a is at level 0. Vertices b , j , and k are at level 1. Vertices c , e , f , and l are at level 2. Vertices d , g , i , m , and n are at level 3. Finally, vertex h is at level 4. Because the largest level of any vertex is 4, this tree has height 4. ◀

A rooted m -ary tree of height h is **balanced** if all leaves are at levels h or $h - 1$.

EXAMPLE 11 Which of the rooted trees shown in Figure 14 are balanced?

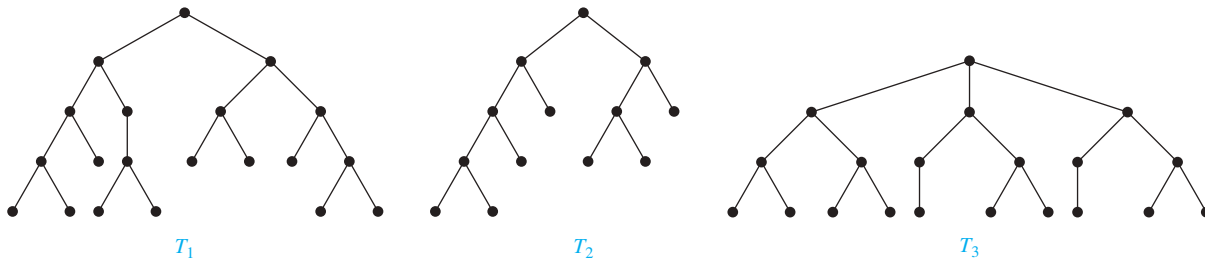


FIGURE 15 Some rooted trees.

Solution: T_1 is balanced, because all its leaves are at levels 3 and 4. However, T_2 is not balanced, because it has leaves at levels 2, 3, and 4. Finally, T_3 is balanced, because all its leaves are at level 3. ◀

A BOUND FOR THE NUMBER OF LEAVES IN AN m -ARY TREE It is often useful to have an upper bound for the number of leaves in an m -ary tree. Theorem 5 provides such a bound in terms of the height of the m -ary tree.

THEOREM 5 There are at most m^h leaves in an m -ary tree of height h .

Proof: The proof uses mathematical induction on the height. First, consider m -ary trees of height 1. These trees consist of a root with no more than m children, each of which is a leaf. Hence, there are no more than $m^1 = m$ leaves in an m -ary tree of height 1. This is the basis step of the inductive argument.

Now assume that the result is true for all m -ary trees of height less than h ; this is the inductive hypothesis. Let T be an m -ary tree of height h . The leaves of T are the leaves of the subtrees of T obtained by deleting the edges from the root to each of the vertices at level 1, as shown in Figure 15.

Each of these subtrees has height less than or equal to $h - 1$. So by the inductive hypothesis, each of these rooted trees has at most m^{h-1} leaves. Because there are at most m such subtrees, each with a maximum of m^{h-1} leaves, there are at most $m \cdot m^{h-1} = m^h$ leaves in the rooted tree. This finishes the inductive argument. ◀

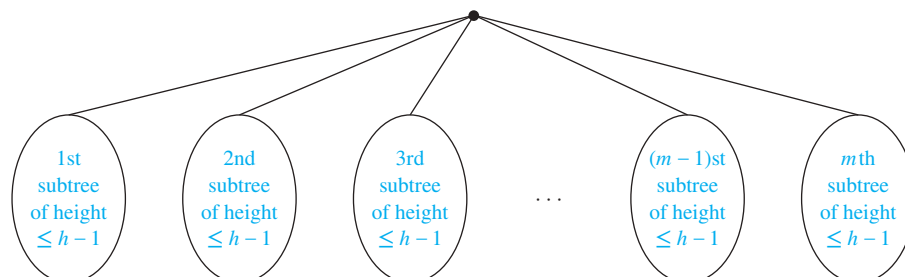


FIGURE 16 The inductive step of the proof.

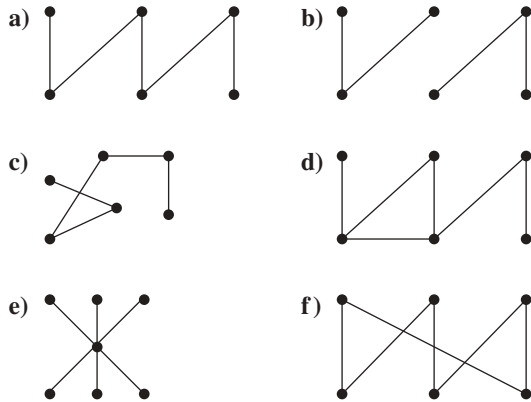
COROLLARY 1

If an m -ary tree of height h has l leaves, then $h \geq \lceil \log_m l \rceil$. If the m -ary tree is full and balanced, then $h = \lceil \log_m l \rceil$. (We are using the ceiling function here. Recall that $\lceil x \rceil$ is the smallest integer greater than or equal to x .)

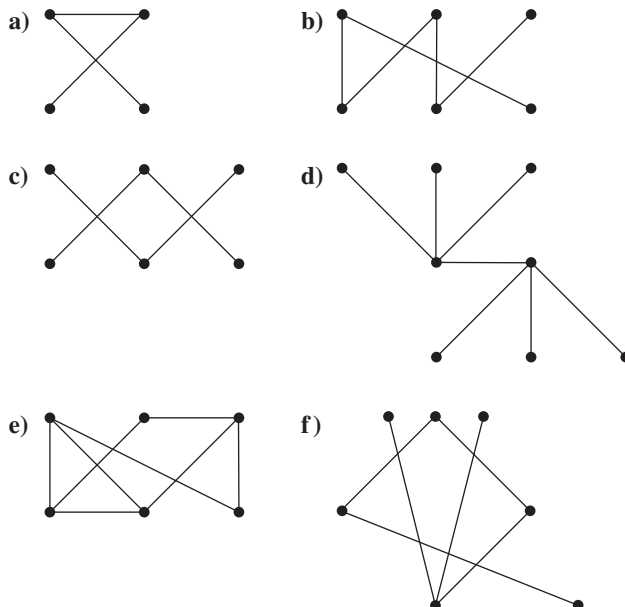
Proof: We know that $l \leq m^h$ from Theorem 5. Taking logarithms to the base m shows that $\log_m l \leq h$. Because h is an integer, we have $h \geq \lceil \log_m l \rceil$. Now suppose that the tree is balanced. Then each leaf is at level h or $h - 1$, and because the height is h , there is at least one leaf at level h . It follows that there must be more than m^{h-1} leaves (see Exercise 30). Because $l \leq m^h$, we have $m^{h-1} < l \leq m^h$. Taking logarithms to the base m in this inequality gives $h - 1 < \log_m l \leq h$. Hence, $h = \lceil \log_m l \rceil$. $\triangleleft$

Exercises

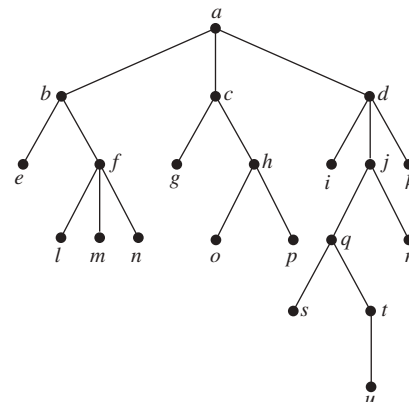
1. Which of these graphs are trees?



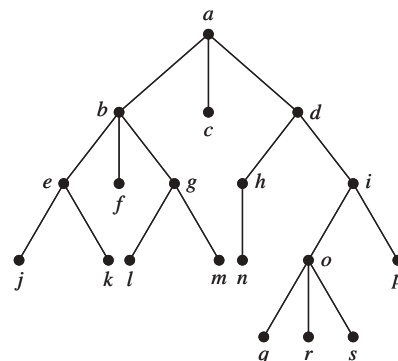
2. Which of these graphs are trees?



3. Answer these questions about the rooted tree illustrated.



- Which vertex is the root?
 - Which vertices are internal?
 - Which vertices are leaves?
 - Which vertices are children of j ?
 - Which vertex is the parent of h ?
 - Which vertices are siblings of o ?
 - Which vertices are ancestors of m ?
 - Which vertices are descendants of b ?
4. Answer the same questions as listed in Exercise 3 for the rooted tree illustrated.



5. Is the rooted tree in Exercise 3 a full m -ary tree for some positive integer m ?
 6. Is the rooted tree in Exercise 4 a full m -ary tree for some positive integer m ?
 7. What is the level of each vertex of the rooted tree in Exercise 3?
 8. What is the level of each vertex of the rooted tree in Exercise 4?
 9. Draw the subtree of the tree in Exercise 3 that is rooted at
 - a) a .
 - b) c .
 - c) e .
 10. Draw the subtree of the tree in Exercise 4 that is rooted at
 - a) a .
 - b) c .
 - c) e .
 11. a) How many nonisomorphic unrooted trees are there with three vertices?
 b) How many nonisomorphic rooted trees are there with three vertices (using isomorphism for directed graphs)?
 - *12. a) How many nonisomorphic unrooted trees are there with four vertices?
 b) How many nonisomorphic rooted trees are there with four vertices (using isomorphism for directed graphs)?
 - *13. a) How many nonisomorphic unrooted trees are there with five vertices?
 b) How many nonisomorphic rooted trees are there with five vertices (using isomorphism for directed graphs)?
 - *14. Show that a simple graph is a tree if and only if it is connected but the deletion of any of its edges produces a graph that is not connected.
 -  *15. Let G be a simple graph with n vertices. Show that
 - a) G is a tree if and only if it is connected and has $n - 1$ edges.
 - b) G is a tree if and only if G has no simple circuits and has $n - 1$ edges. [Hint: To show that G is connected if it has no simple circuits and $n - 1$ edges, show that G cannot have more than one connected component.]
 16. Which complete bipartite graphs $K_{m,n}$, where m and n are positive integers, are trees?
 17. How many edges does a tree with 10,000 vertices have?
 18. How many vertices does a full 5-ary tree with 100 internal vertices have?
 19. How many edges does a full binary tree with 1000 internal vertices have?
 20. How many leaves does a full 3-ary tree with 100 vertices have?
 21. Suppose 1000 people enter a chess tournament. Use a rooted tree model of the tournament to determine how many games must be played to determine a champion, if a player is eliminated after one loss and games are played until only one entrant has not lost. (Assume there are no ties.)
 22. A chain letter starts when a person sends a letter to five others. Each person who receives the letter either sends it to five other people who have never received it or does not send it to anyone. Suppose that 10,000 people send out the letter before the chain ends and that no one receives more than one letter. How many people receive the letter, and how many do not send it out?
 23. A chain letter starts with a person sending a letter out to 10 others. Each person is asked to send the letter out to 10 others, and each letter contains a list of the previous six people in the chain. Unless there are fewer than six names in the list, each person sends one dollar to the first person in this list, removes the name of this person from the list, moves up each of the other five names one position, and inserts his or her name at the end of this list. If no person breaks the chain and no one receives more than one letter, how much money will a person in the chain ultimately receive?
 - *24. Either draw a full m -ary tree with 76 leaves and height 3, where m is a positive integer, or show that no such tree exists.
 - *25. Either draw a full m -ary tree with 84 leaves and height 3, where m is a positive integer, or show that no such tree exists.
 - *26. A full m -ary tree T has 81 leaves and height 4.
 - a) Give the upper and lower bounds for m .
 - b) What is m if T is also balanced?
- A **complete m -ary tree** is a full m -ary tree in which every leaf is at the same level.
27. Construct a complete binary tree of height 4 and a complete 3-ary tree of height 3.
 28. How many vertices and how many leaves does a complete m -ary tree of height h have?
 29. Prove
 - a) part (ii) of Theorem 4.
 - b) part (iii) of Theorem 4.
 -  30. Show that a full m -ary balanced tree of height h has more than m^{h-1} leaves.
 31. How many edges are there in a forest of t trees containing a total of n vertices?
 32. Explain how a tree can be used to represent the table of contents of a book organized into chapters, where each chapter is organized into sections, and each section is organized into subsections.
 33. How many different isomers do these saturated hydrocarbons have?
 - a) C_3H_8
 - b) C_5H_{12}
 - c) C_6H_{14}
 34. What does each of these represent in a tree that represents the structure of an organization?
 - a) the parent of a vertex
 - b) a child of a vertex
 - c) a sibling of a vertex
 - d) the ancestors of a vertex
 - e) the descendants of a vertex
 - f) the level of a vertex
 - g) the height of the tree